

Heinrich Weber, *Lehrbuch der Algebra*, Volume II
English Translation Draft – Batch 13
Sections 58–69; Source pp. 220–271

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§58. The Icosahedral Group

Since in the icosahedral group there is only one system of conjugate fivefold poles, we may, by the preceding reduction, assume that the group contains the two substitutions

$$\Theta(x) = \varepsilon x, \quad \psi(x) = -\frac{1}{x}, \quad (1)$$

where ε is a primitive fifth root of unity. We choose here, as we are free to do, the second substitution in the form $-1 : x$, because the formulae thereby become simpler.

The basic form of degree twelve belonging to the system of fivefold poles must, since it admits the substitutions (1), have the form

$$f(x_1, x_2) = x_1 x_2 (x_1^{10} + m x_1^5 x_2^5 - x_2^{10}), \quad (2)$$

and it remains to determine the constant factor m . The two other basic forms have degrees twenty and thirty. We can determine m from the theorem of §54, if we can construct from f a covariant of degree less than sixty which cannot be represented as a product of powers of three forms of degrees twelve, twenty, and thirty. Such a covariant must then vanish identically.

A covariant of degree sixteen of the form f is easily obtained by taking, in the sense of the first volume, the fourth polar of the binary form $f(x_1, x_2)$:

$$12 \cdot 11 \cdot 10 \cdot 9 P_4(x, \xi) = u_0 \xi_1^4 + 4u_1 \xi_1^3 \xi_2 + 6u_2 \xi_1^2 \xi_2^2 + 4u_3 \xi_1 \xi_2^3 + u_4 \xi_2^4, \quad (3)$$

where the u_i are obtained from the fourth partial derivatives of f . From this fourth polar one forms, as in the theory of binary quartics, its first invariant,

$$u_0 u_4 - 4u_1 u_3 + 3u_2^2. \quad (4)$$

Since a form of degree sixteen cannot be a product of forms of degrees twelve, twenty, and thirty, this covariant must vanish identically.

For the form (2) one has, up to the common numerical factor coming from the fourth differentiation,

$$\begin{aligned} u_0 &= 11 \cdot 10 \cdot 9 \cdot 8 x_1^7 x_2 + 6 \cdot 5 \cdot 4 \cdot 3 m x_1^2 x_2^6, \\ u_1 &= 4 \cdot 11 \cdot 10 \cdot 9 x_1^6 x_2^2 + 4 \cdot 6 \cdot 5 \cdot 4 m x_1 x_2^7, \\ u_2 &= 6 \cdot 5 \cdot m x_1^5 x_2^5, \\ u_3 &= -4 \cdot 11 \cdot 10 \cdot 9 x_1^2 x_2^6 + 4 \cdot 6 \cdot 5 \cdot 4 m x_1^7 x_2, \\ u_4 &= -11 \cdot 10 \cdot 9 \cdot 8 x_1 x_2^7 + 6 \cdot 5 \cdot 4 \cdot 3 m x_1^6 x_2^2. \end{aligned}$$

Substitution in (4), and comparison of the coefficient of the appropriate middle term, gives

$$m = \pm 11.$$

Both signs are admissible. The one choice is carried into the other by replacing x_1 by $-x_1$, or equivalently by an allowable transformation of the group. We therefore put

$$f(x_1, x_2) = x_1 x_2 (x_1^{10} + 11 x_1^5 x_2^5 - x_2^{10}). \quad (5)$$

The ten remaining fivefold poles are consequently obtained by solving

$$x^{10} + 11x^5 - 1 = 0.$$

Writing

$$\omega = -\frac{11 + 5\sqrt{5}}{2}, \quad \omega' = -\frac{11 - 5\sqrt{5}}{2}, \quad (6)$$

so that

$$\omega^2 + \omega = 1, \quad \omega + \omega' = -1, \quad \omega\omega' = -1, \quad (7)$$

these fivefold poles, besides 0 and ∞ , may be written in the form

$$x = \varepsilon^\lambda \omega, \quad x = \varepsilon^\lambda \omega', \quad \lambda = 0, 1, 2, 3, 4. \quad (8)$$

Here $\varepsilon + \varepsilon^{-1} = 2 \cos(2\pi/5) = (-1 + \sqrt{5})/2$ and $\varepsilon^2 + \varepsilon^{-2} = 2 \cos(4\pi/5) = (-1 - \sqrt{5})/2$, so that the notation is compatible with the elementary identities just displayed.

We now apply the theorem of §53. If a, a' are the two poles of a substitution Θ , and if b is a pole conjugate with a , then there is a substitution χ of the group such that $b = \chi(a)$, and such that $b' = \chi(a')$ is the second pole of $\chi^{-1}\Theta\chi$. We may take $a = \infty$, $a' = 0$, and $b = \omega$. Then b' is a still undetermined pole ξ among those in (8), and χ has the form

$$\chi(x) = \frac{\omega\alpha x + \beta}{\alpha x + \beta\omega^{-1}}, \quad \alpha\beta(\omega^{-1} - \omega) = 1. \quad (9)$$

For every exponent h not divisible by five, the substitution Θ^h may be used. The poles of $\chi^{-1}\Theta^h\chi$ must be ω and ξ . From the corresponding quadratic equation for the fixed points one obtains

$$\frac{\beta}{\alpha} = \pm\omega\sqrt{-\xi\omega^{-1}},$$

and the condition (9) then determines the remaining scalar factor. In Weber's normalization this reduces to the simple form

$$\chi(x) = \frac{\omega x + \beta}{x - \omega^{-1}}, \quad (10)$$

where β is still to be determined and, from the determinant condition, has the value $-\omega\xi$.

If χ is known, the whole icosahedral group can be generated. Let r, s run through the residues 0, 1, 2, 3, 4. The group then contains the substitutions

$$\Theta^r, \quad \psi\Theta^r, \quad \Theta^r\chi\Theta^s, \quad \Theta^r\chi\psi\Theta^s. \quad (11)$$

There are sixty of them. Written as matrices of fractional-linear substitutions, they are

$$\Theta^r = \begin{pmatrix} \varepsilon^r & 0 \\ 0 & 1 \end{pmatrix}, \quad (12)$$

$$\psi\Theta^r = \begin{pmatrix} 0 & 1 \\ -\varepsilon^r & 0 \end{pmatrix}, \quad (13)$$

$$\Theta^r\chi\Theta^s = \begin{pmatrix} \varepsilon^r\omega & \varepsilon^{r-s}\beta \\ 1 & -\varepsilon^{-s}\omega \end{pmatrix}, \quad (14)$$

$$\Theta^r\chi\psi\Theta^s = \begin{pmatrix} -\varepsilon^{r+s}\beta & \varepsilon^r\omega \\ \varepsilon^s\omega & 1 \end{pmatrix} = \begin{pmatrix} -\varepsilon^r\beta\omega^{-1} & \varepsilon^{r-s} \\ 1 & \varepsilon^{-s}\omega^{-1} \end{pmatrix}. \quad (15)$$

Since $-\omega^2$ is no power of ε , no two of these matrices represent the same substitution.

The icosahedral group has, as we have seen, twelve fivefold poles. Any two of these are the poles of four substitutions of fifth order; together with the identity these form a cycle of five elements. Thus there are twenty-four elements of fifth order. Similarly, the twenty threefold poles lead to twenty substitutions of third order, and the thirty twofold poles, paired two by two, lead to fifteen substitutions of second order. Together with the identity this gives

$$24 + 20 + 15 + 1 = 60.$$

The determination of β in (10) is obtained by considering the orders of the substitutions (14) and (15). For an arbitrary linear substitution

$$S = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

the second and third iterates are

$$S^2 = \begin{pmatrix} a^2 + bc & b(a + d) \\ c(a + d) & bc + d^2 \end{pmatrix},$$

$$S^3 = \begin{pmatrix} a^3 + 2abc + bcd & b(a^2 + bc + ad + d^2) \\ c(a^2 + bc + ad + d^2) & d^3 + abc + 2bcd \end{pmatrix}.$$

Disregarding the cases where b or c vanishes, which do not occur here, one obtains the necessary and sufficient conditions

$$a + d = 0 \tag{16}$$

for a substitution of second order, and

$$a^2 + bc + ad + d^2 = 0 \tag{17}$$

for one of third order.

The substitutions (12), apart from the identity, have fifth order. The five substitutions (13) have second order. Among the substitutions (14), exactly the five of the form $\Theta^r \chi \Theta^{-r}$ have second order. Hence five among the substitutions (15) must also have second order. From (16) this is possible only if β is a power of ε .

The substitutions of third order must now be distributed among the forms (14) and (15). The condition that (14) have third order is

$$\omega^2(\varepsilon^r + \varepsilon^{-s} + \varepsilon^{-(r+s)}) = 1, \tag{18}$$

and for (15) the condition is

$$\omega^2 = \beta - \varepsilon^{-(r+s)} - \beta^2 \varepsilon^{r+s}. \tag{19}$$

Either one of these conditions can be satisfied only when

$$\beta = 1.$$

Indeed, in (18) the left side is real only when β is real; since β is a power of ε , this forces $\beta = 1$. In (19) one obtains the same conclusion by passing to conjugate-imaginary quantities; the expression can remain invariant under conjugation only if $\beta = \beta^{-1}$, hence $\beta = 1$ among fifth roots of unity.

With $\beta = 1$, five of the substitutions (15), namely those of the form $\Theta^r \chi \psi \Theta^{-r}$, have second order. The substitutions (14) have third order precisely when

$$r + s \equiv \pm 2 \pmod{5}, \tag{20}$$

and the substitutions (15) have third order precisely when

$$r + s \equiv \pm 1 \pmod{5}. \tag{21}$$

Thus exactly ten substitutions of second order and twenty of third order occur in the forms (14) and (15); the remaining cases are of fifth order.

Summarizing, χ must have the form

$$\chi(x) = \frac{\omega x + 1}{x - \omega^{-1}}. \tag{22}$$

Among the sixty substitutions

$$\Theta^r, \quad \psi \Theta^r, \quad \Theta^r \chi \Theta^s, \quad \Theta^r \chi \psi \Theta^s, \tag{23}$$

there occur, besides the identity,

1. fifteen substitutions of second order:

$$\psi\Theta^r, \quad \Theta^r\chi\Theta^{-r}, \quad \Theta^r\chi\psi\Theta^{-r} \quad (r = 0, 1, 2, 3, 4);$$

2. twenty substitutions of third order:

$$\Theta^r\chi\Theta^s \quad (r + s \equiv \pm 2), \quad \Theta^r\chi\psi\Theta^s \quad (r + s \equiv \pm 1);$$

3. twenty-four substitutions of fifth order:

$$\Theta^r \quad (r \neq 0), \quad \Theta^r\chi\Theta^s \quad (r + s \equiv \pm 1), \quad \Theta^r\chi\psi\Theta^s \quad (r + s \equiv \pm 2).$$

In explicit fractional-linear form these substitutions include

$$\varepsilon^r x, \quad -\frac{1}{\varepsilon^r x}, \quad \frac{\varepsilon^r \omega x + \varepsilon^{r-s}}{x - \varepsilon^{-s} \omega}, \quad \frac{-\varepsilon^r \omega^{-1} x + \varepsilon^{r-s}}{x + \varepsilon^{-s} \omega^{-1}}. \quad (24)$$

It remains only to verify that the totality of substitutions (23) is indeed a group. This follows from the composition laws, which we now record. First, directly from the meaning of Θ, ψ, χ ,

$$\Theta^5 = 1, \quad \psi^2 = 1, \quad \psi\Theta = \Theta^{-1}\psi, \quad \chi\psi = \psi\chi. \quad (25)$$

For an arbitrary exponent λ , use

$$\omega = \varepsilon + \varepsilon^{-1}, \quad \omega' = \varepsilon^2 + \varepsilon^{-2}, \quad \omega\omega' = -1. \quad (26)$$

Then, putting first $\lambda = 1$ and then $\lambda = 2$, Weber obtains

$$\chi\Theta^{\pm 1}\chi = \Theta^{\pm 1}\chi\psi\Theta^{\pm 1}, \quad \chi\Theta^{\pm 2}\chi = \Theta^{\pm 2}\chi\Theta^{\pm 2}. \quad (27)$$

Together with (25), these formulae allow any product of two substitutions of the form (23) to be reduced again to one of the same forms. Thus the group property is proved.

From the first formula in (27) one also derives

$$\psi = \Theta\chi^{-1}\Theta^{-1}\chi^{-1}\Theta,$$

so that ψ may be derived from Θ and χ . The substitutions Θ and χ alone may therefore be taken as generators of the icosahedral group.

To write χ with determinant one, we use

$$-\omega^{-2} - 1 = -\omega^{-1} = (\varepsilon - \varepsilon^{-1})^2.$$

Replacing ω by $\varepsilon + \varepsilon^{-1}$, this gives

$$\chi = \frac{1}{\varepsilon - \varepsilon^{-1}} \begin{pmatrix} \omega & 1 \\ 1 & -\omega^{-1} \end{pmatrix} = \begin{pmatrix} \frac{\omega}{\varepsilon - \varepsilon^{-1}} & \frac{1}{\varepsilon - \varepsilon^{-1}} \\ \frac{1}{\varepsilon - \varepsilon^{-1}} & -\frac{\omega^{-1}}{\varepsilon - \varepsilon^{-1}} \end{pmatrix}. \quad (28)$$

§59. Divisors of the Icosahedral Group

The divisors, or subgroups, of the icosahedral group must be sought among the lower polyhedral groups. First there are cyclic groups C_n , each generated by one of the icosahedral substitutions.

Counting them immediately from the preceding arrangement according to the orders of the substitutions gives

$$15 \text{ groups } C_2, \quad 10 \text{ groups } C_3, \quad 6 \text{ groups } C_5.$$

There are also dihedral groups D_2, D_3, D_5 in the icosahedral group. Representatives may be written in the form

$$\begin{aligned} D_2 : & 1, \chi\psi, \Theta^{-1}\chi\psi\Theta^{-1}, \Theta\chi\psi\Theta, \\ D_3 : & \Theta^r\chi\Theta^s \quad (r+s \equiv \pm 2 \pmod{5}), \\ D_5 : & \Theta^r, \psi\Theta^r, \quad r = 0, 1, 2, 3, 4. \end{aligned}$$

The number of such dihedral groups is found from the fact that a dihedral group is completely determined by its cyclic subgroup. For the four-group D_2 one must remember that the same group is obtained whether one begins with ψ , $\chi\psi$, or $\Theta\chi\psi\Theta$; hence the directly obtained number is to be divided by three. Thus the numbers of groups D_2, D_3, D_5 are respectively

$$5, \quad 10, \quad 6.$$

Of special importance are the tetrahedral groups contained in the icosahedral group. We determine one of them, denoted by Q . A tetrahedral group must contain a four-group D_2 ; we take the one generated by ψ and $\chi\psi$. In addition, Q must contain a substitution of third order. Such a substitution can be taken in one of the two forms

$$\Theta^r\chi\Theta^s, \quad \Theta^r\chi\psi\Theta^s.$$

Both choices lead to the same result; choose

$$\varphi = \Theta^r\chi\Theta^s, \quad r+s \equiv \pm 2 \pmod{5}. \quad (29)$$

Neither of the numbers r, s may be zero; otherwise either $\chi\varphi$ or $\varphi\chi$ would be a power of Θ , hence of fifth order, while no element of fifth order can occur in Q .

Using the composition laws of §58 one obtains

$$\chi\varphi = \Theta^r\chi\psi\Theta^{r+s} \quad (r \neq \pm 1), \quad \chi\varphi = \Theta^{-r}\chi\Theta^{-r+s} \quad (r \neq \pm 2),$$

with the exclusions required by the same reason just given. The four possible cases left are

$$r = \pm 1, \quad s = \pm 2, \quad \text{or} \quad r = \pm 2, \quad s = \pm 1 \pmod{5}.$$

If one of the four corresponding substitutions of third order belongs to Q , the other three do also. Put

$$\varphi = \Theta\chi\Theta. \quad (30)$$

Then from the relations of §58 one obtains

$$\varphi^2 = \Theta^{-1}\chi\Theta^{-1}, \quad \varphi\chi = \Theta^{-1}\chi\psi\Theta^{-2}, \quad \chi\varphi = \Theta^2\chi\psi\Theta, \quad (31)$$

and, by adjoining the compositions with ψ , one sees that the same group is obtained if one begins with a substitution of the second form.

The whole group Q is therefore completely determined by the chosen four-group D_2 . It contains the elements

$$1, \quad \psi, \quad \chi\psi, \quad \Theta^{-1}\chi\psi\Theta^{-1}, \quad \varphi, \quad \varphi\psi, \quad \varphi\chi, \quad \varphi\chi\psi, \quad \varphi^2, \quad \varphi^2\psi, \quad \varphi^2\chi, \quad \varphi^2\chi\psi. \quad (32)$$

Equivalently, Q is generated by φ, ψ, χ and is represented in the form of a tetrahedral group. The multiplication rules following from (30) are

$$\psi\varphi^{-1} = \chi\varphi^{-1}\psi, \quad \chi\varphi = \varphi\chi\psi, \quad \psi\chi = \chi\psi, \quad \chi^2 = \psi^2 = 1. \quad (33)$$

They are the defining composition laws of the tetrahedral group.

Since the tetrahedral group contains five four-groups D_2 , the icosahedral group contains five tetrahedral subgroups. They are obtained from Q by transforming with the powers of Θ :

$$\Theta^{-r}Q\Theta^r, \quad r = 0, 1, 2, 3, 4.$$

No two of these groups have, besides the identity, a common substitution. Indeed, Q and $\Theta^{-1}Q\Theta$ have in common, besides the identity, only two displayed elements, and neither of these occurs in $Q\Theta^{-1}$.

It follows from the theorem on permutation representations by cosets that the icosahedral group is isomorphic to a permutation group of degree sixty on five letters. The representation is obtained by adjoining to the five right cosets

$$Q, \quad Q\Theta, \quad Q\Theta^2, \quad Q\Theta^3, \quad Q\Theta^4$$

each element δ of the icosahedral group, forming

$$Q\delta, \quad Q\Theta\delta, \quad Q\Theta^2\delta, \quad Q\Theta^3\delta, \quad Q\Theta^4\delta,$$

and examining the permutation so produced. For the generating substitutions $\delta = \Theta, \chi$, one obtains respectively the permutations

$$(Q, Q\Theta, Q\Theta^2, Q\Theta^3, Q\Theta^4) \mapsto (Q\Theta, Q\Theta^2, Q\Theta^3, Q\Theta^4, Q),$$

and

$$(Q, Q\Theta, Q\Theta^2, Q\Theta^3, Q\Theta^4) \mapsto (Q, Q\Theta^2, Q\Theta^4, Q\Theta, Q\Theta^3).$$

The second computation follows from the formulae

$$\Theta\chi = \varphi\Theta^{-1}, \quad \Theta^2\chi = \chi\varphi^2\Theta^3, \quad \Theta^3\chi = \varphi^2\Theta, \quad \Theta^4\chi = \varphi\chi\Theta^2.$$

These permutations are all even; hence the permutation group is the alternating group on five letters. Consequently the icosahedral group is simple, and coincides with the group provisionally called the icosahedral group in the earlier abstract discussion.

§60. The Basic Forms of the Icosahedral Group

We have already found one of the three basic forms of the icosahedral group, namely

$$f = x_1x_2(x_1^{10} + 11x_1^5x_2^5 - x_2^{10}). \quad (34)$$

The two missing forms have degrees twenty and thirty.

The basic form of degree twenty is the Hessian covariant of f . We put

$$H = \frac{1}{121} \left(f''(x_1, x_1)f''(x_2, x_2) - f''(x_1, x_2)^2 \right). \quad (35)$$

A straightforward calculation gives

$$H = -(x_1^{20} + x_2^{20}) + 228(x_1^{15}x_2^5 - x_1^5x_2^{15}) - 494x_1^{10}x_2^{10}. \quad (36)$$

The basic form of degree thirty may be defined as the functional determinant of H and f . Set

$$T = \frac{1}{20} (f'(x_1)H'(x_2) - f'(x_2)H'(x_1)). \quad (37)$$

Then

$$T = (x_1^{30} + x_2^{30}) + 522(x_1^{25}x_2^5 - x_1^5x_2^{25}) - 10005(x_1^{20}x_2^{10} + x_1^{10}x_2^{20}). \quad (38)$$

These three forms satisfy an identical relation. To find it, put

$$\lambda = x_1^{10} + x_2^{10}, \quad \mu = x_1^5x_2^5.$$

Then

$$f^5 = \mu(\lambda + 11\mu)^5,$$

and after extracting the common factor $x_1^{10} + x_2^{10}$ from T and forming T^2 , a numerical calculation gives

$$T^2 + H^3 = 1728f^5. \quad (39)$$

This is the required relation.

The icosahedral group is simple, hence by the preceding theory it has no relative invariants. Thus the forms f, H, T remain absolutely unchanged when subjected to the binary icosahedral substitutions written with determinant one. This can also be checked directly, without using the general theorem.

Relation (39) may be written as

$$\frac{T^2}{f^5} + \frac{H^3}{f^5} = 1728. \quad (40)$$

The two quotients T^2/f^5 and H^3/f^5 may be regarded as functions of the single variable $x = x_1 : x_2$. If an icosahedral substitution is applied to this variable, the quotients can change only by constant factors. If

$$\frac{T^2}{f^5}(y) = h \frac{T^2}{f^5}(x), \quad \frac{H^3}{f^5}(y) = k \frac{H^3}{f^5}(x),$$

then, because both x and y are independent variables, the identity (40) gives

$$h \frac{T^2}{f^5} + k \frac{H^3}{f^5} = 1728.$$

Since the two quotients are not in a constant ratio, one must have $h = k = 1$. Hence the quotients remain absolutely unchanged. It follows in turn that f, H, T themselves remain absolutely unchanged under homogeneous icosahedral substitutions of determinant one. For if f were changed into cf , then H and T would be changed into c^2H and c^3T ; since the quotients are unchanged, $c = 1$.

If we set

$$\frac{H^3}{f^5} = z,$$

then (39) gives

$$H^3 - zf^5 = 0, \quad (41)$$

and equivalently

$$T^2 - (1728 - z)f^5 = 0. \quad (42)$$

These are two different forms of the same equation of degree sixty in x , called the *icosahedral equation*. Its properties and its relation to the general equation of the fifth degree will be treated later.

§61. The Invariants of the Icosahedron

The basic forms of the polyhedral groups exhaust, as we now show, the invariant theory of the corresponding groups. We give the proof only for the icosahedral group. The proofs for the other groups are entirely analogous and may be supplied by the reader. The fact that, in the tetrahedral and octahedral groups, relative invariants occur besides absolute invariants, whereas the icosahedral group, being simple, has only absolute invariants, has no essential influence on the argument.

We prove that every invariant of the icosahedral group can be represented as an integral rational function of the three forms f, H, T .

First:

1. No two of the icosahedral forms f, H, T have a common divisor.

This follows at once from the definition of these functions: the equations $f = 0$, $H = 0$, and $T = 0$ give respectively the fivefold, threefold, and twofold poles, and these poles are all distinct.

2. A double root of the icosahedral equation

$$H^3 - zf^5 = 0 \tag{43}$$

is necessarily a root of f , H , or T . Hence it can occur only for one of the values $z = 0, \infty, 1728$.

Indeed, if (43) has a double root, then the function and its first derivative vanish together. In homogeneous notation, by Euler's theorem, the two partial derivatives with respect to x_1 and x_2 vanish simultaneously:

$$\begin{aligned} 3H^2H'(x_1) - 5zf^4f'(x_1) &= 0, \\ 3H^2H'(x_2) - 5zf^4f'(x_2) &= 0. \end{aligned}$$

Since H and f cannot vanish simultaneously, there are only three possibilities: $H = 0$, or $f = 0$, or

$$H'(x_1)f'(x_2) - H'(x_2)f'(x_1) = 0,$$

i.e. $T = 0$, which corresponds by §60 to $z = 1728$.

3. If $J(x_1, x_2)$ is any invariant form of the icosahedral group and ξ is one of its roots, so that $J(\xi, 1) = 0$, then all values obtained from ξ by the fractional icosahedral substitutions are also roots of J .

Consequently, if J has a common divisor with one of the functions f, H, T , then J is divisible by that corresponding form.

Remove from J the highest possible powers of the three basic forms f, H, T , so that the remaining invariant is relatively prime to all three. If ξ is then a root of the remaining form, choose z so that ξ is also a root of (43). By item 2, ξ is a simple root of that equation. By item 3 all of its icosahedral transforms are roots of J , so J is divisible by the left side of (43). The quotient is again an invariant form, and the argument repeats.

We arrive at the theorem:

4. Every invariant of the icosahedron can be represented in the form

$$J(x_1, x_2) = Cf^\alpha H^\beta T^\gamma F(H^3, f^5), \tag{44}$$

where α, β, γ are non-negative integers, C is a constant, and F is a homogeneous integral function.

§62. Polyhedral Groups of the Second Kind. Crystallographic Groups

In §51 we already drew attention to groups of linear ternary substitutions which contain not only proper but also improper orthogonal substitutions. We called them groups of the second kind. We shall now consider the finite ones among them as polyhedral groups of the second kind, or extended polyhedral groups. The substitutions in them with determinant -1 will be called substitutions of the second kind.

According to §50 these groups are isomorphic to groups of binary linear substitutions with determinant $+1$ and -1 . For the fractional substitutions which led us to the polyhedral groups, however, the two kinds cannot be distinguished.

The finite groups of the second kind are important, apart from their general group-theoretic interest, because of their application to crystallography. Their complete determination now presents no essential difficulty. By substitutions we shall here simply mean binary linear substitutions of determinant ± 1 , remembering that two such substitutions whose coefficients differ only by a common sign, such as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \begin{pmatrix} -a & -b \\ -c & -d \end{pmatrix},$$

are not to be considered distinct.

Every group of linear substitutions contains the identity, whose determinant is $+1$. Therefore every group Q of the second kind contains, besides improper substitutions, proper substitutions; and these proper substitutions themselves form a group P . If φ is any substitution of the second kind contained in Q , then the product of φ with any other substitution of the second kind in Q lies in P . Hence

$$Q = P + P\varphi. \quad (45)$$

The group P must be one of the polyhedral groups considered above. Moreover

$$\varphi^{-1}P\varphi = P,$$

so that P is a normal divisor of Q . Conversely, if P is a polyhedral group and φ is a substitution of the second kind satisfying $\varphi^{-1}P\varphi = P$, then $P + P\varphi$ is a group of the second kind.

As before, two groups which are transformed into one another are not considered essentially different. We must run through the cases.

I. The identity group

If P is the identity group, viewed as the cyclic group of first degree C_1 , then P contains only the identity. Hence $\varphi^2 = 1$, which is also sufficient. By transformation φ may be brought into the form

$$\begin{pmatrix} a & b \\ c & -a \end{pmatrix},$$

with the condition $a^2 + bc = 1$. This gives two essentially different forms:

$$\varphi' = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \varphi'' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (46)$$

The first is called inversion, the second reflection. Translated into rectangular coordinates by §50, φ' changes the signs of all three coordinates, whereas φ'' changes x into $-x$. Thus the two groups of the second kind are

$$C'_1 = C_1 + C_1\varphi', \quad C''_1 = C_1 + C_1\varphi''.$$

II. Cyclic groups

Let $P = C_n$, $n > 1$. Put

$$c = \begin{pmatrix} \varepsilon & 0 \\ 0 & \varepsilon^{-1} \end{pmatrix}, \quad \varepsilon = e^{\pi i/n}. \quad (47)$$

Then, as in §55,

$$C_n = 1, c, c^2, \dots, c^{n-1}. \quad (48)$$

If

$$\varphi = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad ad - bc = -1, \quad (49)$$

then $\varphi^{-1}c\varphi$ and $\varphi^{-1}d\varphi$ must occur in the row (48). This forces either $b = c = 0$ or $a = d = 0$. In the first case one obtains two possible extensions,

$$C'_n = \left\{ \begin{pmatrix} e^{\pi i h/n} & 0 \\ 0 & e^{-\pi i h/n} \end{pmatrix} : h = 0, 1, \dots, 2n-1 \right\}, \quad (50)$$

and

$$C''_n = \left\{ \pm \begin{pmatrix} e^{\pi i h/n} & 0 \\ 0 & e^{-\pi i h/n} \end{pmatrix} : h = 0, 1, \dots, n-1 \right\}. \quad (51)$$

For even n both C'_1 and C''_1 occur in C'_n ; for odd n , one of them occurs in C'_n and the other in C''_n . In the second case $a = d = 0$, a transformation which leaves P unchanged reduces φ to

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

producing a third cyclic extension.

III. Dihedral groups

Let $P = D_n$, consisting of

$$D_n = 1, c, c^2, \dots, c^{n-1}, d, cd, c^2d, \dots, c^{n-1}d, \quad d = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}. \quad (52)$$

The substitution c has order n , and the substitutions $c^h d$ all have order two. If $n > 2$, the transform $\varphi^{-1}c\varphi$ must again be among the powers of c , so the preceding analysis applies. Since d does not occur in the cyclic extensions just described, the dihedral groups of the second kind have the forms

$$C'_n + C'_n d, \quad C''_n + C''_n d, \quad C'''_n + C'''_n d. \quad (53)$$

Written out, the first two are the families

$$\begin{pmatrix} e^{\pi i h/n} & 0 \\ 0 & e^{-\pi i h/n} \end{pmatrix}, \quad \begin{pmatrix} 0 & e^{\pi i h/n} \\ e^{-\pi i h/n} & 0 \end{pmatrix}, \quad h = 0, 1, \dots, 2n-1,$$

and the analogous signed family with $h = 0, 1, \dots, n-1$. The third coincides with one of these forms according as n is odd or even.

The case $n = 2$ gives only an apparent exception. One might have $\varphi^{-1}c\varphi = d$ or cd . Direct calculation gives two groups

$$1, c, d, cd, \varphi_1, \varphi_1 c, \varphi_1 d, \varphi_1 cd, \quad 1, c, d, cd, \varphi_2, \varphi_2 c, \varphi_2 d, \varphi_2 cd, \quad (54)$$

where

$$\varphi_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \varphi_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} i & i \\ i & -i \end{pmatrix}.$$

The formulae

$$c\varphi_1 = \varphi_1 d, \quad d\varphi_1 = \varphi_1 c, \quad cd\varphi_1 = \varphi_1 dc,$$

and their analogues for φ_2 , show that these groups are transformed forms of D_2 .

For the remaining polyhedral groups of the second kind, two general remarks are useful. The inversion

$$j = \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} \quad (55)$$

is a similarity substitution and therefore commutes with every substitution. Thus from every polyhedral group of the first kind one obtains at least one group of the second kind,

$$P + Pj. \quad (56)$$

By Sylow's theorem, every finite group contains a subgroup whose order is the highest power of two dividing the order of the group. The extended tetrahedral, octahedral, and icosahedral groups have orders 24, 48, 120; hence they must contain divisors of order 8, 16, 8. In the tetrahedral and icosahedral groups there is a four-group D_2 but no substitution of fourth order; in the octahedral group there is a dihedral group D_4 but no substitution of eighth order. Hence one of the extended dihedral groups just considered must occur.

For the tetrahedral group two extensions occur. Besides

$$T' = T + Tj,$$

one obtains another

$$T'' = T + Tj_1,$$

where j_1 is the second extension of the four-group. No further tetrahedral extension occurs, because the tetrahedral group is completely determined by any one four-group contained in it. If an extension of the four-group of one of the exceptional forms (54) is chosen, the resulting tetrahedral group is merely a transform of T'' .

For the octahedral group, the special substitution defining the second tetrahedral extension does not commute with the octahedral group, and consequently the octahedral group gives no further extension beyond inversion:

$$O' = O + Oj.$$

The same is true of the icosahedral group. One may transform the icosahedral group so that one of its four-groups has the simplest form; the calculation is longer but entirely straightforward, and leads to no additional case. Thus one obtains

$$J' = J + Jj.$$

The following groups of the second kind are therefore obtained:

$$C'_1, C''_1, \quad C'_n, C''_n, C'''_n, \quad D'_n, D''_n, D'''_n, \\ T', T'', \quad O', \quad J'.$$

The thirty-two symmetry systems of crystallographers are contained among them. In the crystallographic notation they are the groups

$$\begin{array}{ccccccc} C_1, & C'_1, & C''_1, & D_1, & & & \\ C_2, & C'_2, & C''_2, & D_2, & D'_2, & D''_2 & \\ C_3, & C'_3, & C''_3, & D_3, & D'_3, & D''_3 & \\ C_4, & C'_4, & C''_4, & D_4, & D'_4, & & \\ C_6, & C'_6, & C''_6, & D_6, & D'_6, & & \\ T, & T', & T'', & O, & O'. & & \end{array}$$

The remaining polyhedral groups are excluded in crystallography by the crystallographic law of rational indices.

§63. Function Congruences

From the linear substitutions whose construction and composition we have studied in the preceding sections one can derive another kind of finite group, namely the congruence groups.

These groups are of many-sided interest. They arise first in the theory of elliptic functions, where they appear as Galois groups of modular equations. They are also important for general group theory, because they furnish a method for constructing whole series of simple groups. This is all the more noteworthy since, as we saw in the fourth section, simple groups are rather rare, at least among the lower degrees.

The theory of these congruence groups is not only greatly generalized, but the governing laws appear much more sharply, if one bases it on an extension of the notion of congruence due to Gauss and subsequently developed and applied, especially by Galois, in problems of group theory.

To obtain a basis for the theory, consider an integral polynomial in one variable t ,

$$f(t) = a_0 t^n + a_1 t^{n-1} + \cdots + a_{n-1} t + a_n, \quad (57)$$

with integral coefficients. Choose a prime number p as modulus, and assume that a_0 is not divisible by p .

Two integral functions of t whose corresponding coefficients are congruent modulo p will themselves be called congruent modulo p . The function $f(t)$ is called reducible modulo p if there exist two integral functions $\varphi(t), \psi(t)$, each with at least one term depending on t whose coefficient is not divisible by p , such that

$$f(t) \equiv \varphi(t)\psi(t) \pmod{p}. \quad (58)$$

Terms with coefficients divisible by p may of course be omitted from φ and ψ , and then the degree of their product equals the degree of f . Hence the degrees of φ and ψ are both lower than n . If no such functions exist, $f(t)$ is called irreducible modulo p .

A function irreducible modulo p is certainly irreducible over the field of rational numbers. The converse, however, need not hold. For example, a quadratic function $t^2 - B$ is reducible modulo p when B is a quadratic residue of p , for if $a^2 \equiv B \pmod{p}$, then

$$t^2 - B \equiv (t - a)(t + a) \pmod{p}.$$

On the other hand, $t^2 - N$ is irreducible when N is a non-residue of p , since otherwise $t^2 - N$ would vanish modulo p for some rational integer t .

The functions

$$t^2 + t + 1, \quad t^3 + t + 1$$

for the modulus two give another example. They are irreducible modulo two. If either were reducible, one factor would have to be linear, and there would have to be an integer which, substituted for t , makes the function even. This is impossible, since for $t = 0$ and $t = 1$ both functions take odd values.

Let

$$P = t^n + p_1 t^{n-1} + p_2 t^{n-2} + \cdots + p_{n-1} t + p_n \quad (59)$$

be irreducible modulo p , and for simplicity take the leading coefficient to be one. From any other integral function $F(t)$ one may derive a remainder $\Phi(t)$ of degree lower than n by writing

$$F(t) = QP + \Phi(t), \quad (60)$$

where Q has integral coefficients. We describe the relation of F to Φ as a congruence modulo P .

Thus two functions $F(t), F_1(t)$ which have the same remainder modulo P are called congruent modulo P , and this is expressed by

$$F(t) \equiv F_1(t) \pmod{P}.$$

Equivalently, $F(t) - F_1(t)$ is divisible by P .

If the two functions do not have the same remainder, but have remainders congruent modulo p , then the functions are called congruent modulo the double modulus P, p :

$$F(t) \equiv F_1(t) \pmod{P, p}. \quad (61)$$

For this kind of congruence one has:

1. The product of two functions cannot be congruent to zero unless one of the factors is congruent to zero.

The proof follows from the algorithm for the greatest common divisor. Suppose F_1 has degree lower than P and is not congruent to zero modulo p . One can determine functions $F_2, F_3, \dots$, also of lower degree than P , of decreasing degree, and quotients $Q_1, Q_2, \dots$, such that

$$P = Q_1 F_1 + F_2, \quad F_1 = Q_2 F_2 + F_3, \quad \dots, \quad F_{r-2} = Q_{r-1} F_{r-1} + F_r \pmod{p}, \quad (62)$$

until the last remainder F_r is a constant. This constant cannot be congruent to zero modulo p ; otherwise P would be reducible modulo p . If now $Q F_1 \equiv 0 \pmod{P, p}$, multiplication of the equations (62) by Q gives successively $Q F_2 \equiv 0, \dots, Q F_r \equiv 0$, hence $Q \equiv 0$. Taking F_1, Q to be the remainders of the two factors proves the assertion.

The remainders of all functions modulo P have the form

$$X = X(t) = x_0 + x_1 t + \dots + x_{n-1} t^{n-1}. \quad (63)$$

To obtain one representative from each congruence class modulo P, p , it is enough to let each coefficient $x_0, x_1, \dots, x_{n-1}$ run through the residues $0, 1, \dots, p-1$. Thus there are exactly p^n incongruent functions modulo P, p .

§64. Congruence Fields

It is only a different mode of expression for the preceding considerations to introduce, with Galois, an imaginary number ε satisfying

$$P(\varepsilon) \equiv 0 \pmod{p}. \quad (64)$$

No ordinary rational integer of this sort exists when $n > 1$, for then $P(t)$ would be divisible by $t - \varepsilon$ and so would be reducible. Nevertheless one may use such a symbol in calculation, just as one introduces the imaginary unit $i = \sqrt{-1}$ in ordinary arithmetic. As far as addition, subtraction, and multiplication are concerned, one calculates with it according to the ordinary rules of algebra, using the equation $P(\varepsilon) = 0$ whenever desired. One may even think of ε as an actual root of $P(x) = 0$ in the usual sense.

All numbers obtained in this way can be written in the form

$$\alpha = a_0 + a_1 \varepsilon + \dots + a_{n-1} \varepsilon^{n-1}. \quad (65)$$

Because the modulus p is held fixed, two numbers congruent modulo p will simply be regarded as equal. Strictly speaking, therefore, the objects of the calculation are not the individual numbers but

the classes of numbers congruent to one another. These numbers α are called the Galois imaginaries. The system associated with a fixed ε will be denoted by $\mathfrak{G}$.

We have first:

2. There are exactly p^n different numbers in $\mathfrak{G}$.

From item 1 of the preceding section we also obtain:

3. A product of two or more numbers of $\mathfrak{G}$ is zero if and only if at least one factor is zero.

The whole system $\mathfrak{G}$ is obtained by allowing the rational coefficients $a_0, a_1, \dots, a_{n-1}$ in (65) to run independently through a complete residue system modulo p . It contains the ordinary residue classes $0, 1, \dots, p-1$, and for $n=1$ it is identical with that system.

If $\alpha \neq 0$ is fixed in $\mathfrak{G}$, then $\alpha\xi$ runs through the whole system at the same time as ξ . Indeed, $\alpha\xi = \alpha\xi'$ implies $\xi = \xi'$. Hence:

4. If $\alpha, \beta \in \mathfrak{G}$ and $\alpha \neq 0$, there exists exactly one $\gamma \in \mathfrak{G}$ such that

$$\alpha\gamma = \beta.$$

This proves that division, except by zero, is allowed in $\mathfrak{G}$. We write this γ as β/α .

The system $\mathfrak{G}$ may be called a finite field: it contains only finitely many numbers, and it has the characteristic property of a field, namely the unrestricted performance of the arithmetic operations, except division by zero. We shall call $\mathfrak{G}$ a *congruence field*. The integer n is its degree and p its modulus.

If $\alpha \neq 0$, then $\alpha\xi = \eta$ runs through the non-zero elements as ξ runs through them. Multiplying all such equations gives

$$\alpha^{p^n-1}\Pi = \Pi,$$

where Π denotes the non-zero product of all non-zero elements. Cancelling Π yields Fermat's theorem in this field:

$$\alpha^{p^n-1} = 1. \tag{66}$$

Multiplication by α gives the equivalent form

$$\alpha^{p^n} = \alpha, \tag{67}$$

which also holds for $\alpha = 0$.

If $\alpha \neq 0$, let e be the least positive exponent with

$$\alpha^e = 1. \tag{68}$$

Then the powers $1, \alpha, \alpha^2, \dots, \alpha^{e-1}$ are distinct, and α is said to belong to the exponent e . The exponent e divides $p^n - 1$. For if $\alpha^m = 1$ and $m = qe + r$ with $0 \leq r < e$, then $\alpha^r = 1$, hence $r = 0$. In particular $e \mid p^n - 1$.

If $p^n - 1 = ef$, then α^h belongs to the exponent e precisely when h is relatively prime to e . Thus, if there is any element with exponent e , there are $\varphi(e)$ of them, where φ is Euler's function. Since

$$\sum_{e \mid (p^n-1)} \varphi(e) = p^n - 1, \tag{69}$$

and since every non-zero element belongs to exactly one exponent, it follows that, for every divisor e of $p^n - 1$, there are exactly $\varphi(e)$ elements of exponent e . In particular, there are $\varphi(p^n - 1)$ elements γ whose powers

$$1, \gamma, \gamma^2, \dots, \gamma^{p^n-2} \quad (70)$$

are all distinct. Such elements are called primitive roots of $\mathfrak{G}$.

Among the elements of $\mathfrak{G}$ some are squares, and the others are non-squares. If $p = 2$, then every element is a square, as follows from (67). If p is odd, then one half of the non-zero elements are squares and the other half are non-squares. Indeed, the two opposite elements α and $-\alpha$ are distinct but have the same square, and $\beta^2 = \alpha^2$ implies $\beta = \pm\alpha$. Thus there are exactly $(p^n - 1)/2$ non-zero squares and the same number of non-squares. Counting zero among the squares, one has:

5. If $p = 2$, all elements of $\mathfrak{G}$ are squares. If p is odd, there are $(p^n + 1)/2$ squares and $(p^n - 1)/2$ non-squares.

For odd p , a primitive root is necessarily a non-square, and in the sequence (70) the even powers are the squares and the odd powers the non-squares.

The product and quotient of two squares is again a square. Hence the product of a non-square and a non-zero square is a non-square, and the product of two non-squares is a square. All these assertions also follow immediately from the representation by powers of a primitive root.

We close with a result needed later:

6. Every non-square is the sum of two squares.

Here p is assumed odd, since otherwise there are no non-squares. First, every element β can be represented as a difference of two squares, because

$$\beta = \left(\frac{\beta+1}{2}\right)^2 - \left(\frac{\beta-1}{2}\right)^2.$$

If -1 is a square in $\mathfrak{G}$, this is already a representation of β as a sum of two squares. If -1 is a non-square, choose in the natural sequence $1, 2, \dots, p-1$ a square immediately followed by a non-square. Thus let $a = \xi^2$ be a square and $\xi^2 + 1$ a non-square. If β is a non-square, then $\beta/(\xi^2 + 1) = \eta^2$ is a square, and so

$$\beta = (\xi\eta)^2 + \eta^2.$$

§65. Congruence Groups in the Field $\mathfrak{G}$

In every congruence field $\mathfrak{G}$ one can form a finite group of linear substitutions

$$A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \quad (71)$$

by letting $\alpha, \beta, \gamma, \delta$ run through all elements of $\mathfrak{G}$, and by composing two substitutions according to

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} = \begin{pmatrix} \alpha\alpha' + \beta\gamma' & \alpha\beta' + \beta\delta' \\ \gamma\alpha' + \delta\gamma' & \gamma\beta' + \delta\delta' \end{pmatrix}. \quad (72)$$

The degree of this full group is p^{4n} when p is the modulus and n the degree of $\mathfrak{G}$.

It contains a group of smaller degree. Let

$$\det A = \alpha\delta - \beta\gamma.$$

By (72), determinants multiply. Therefore, if the entries are restricted by

$$\alpha\delta - \beta\gamma = 1, \quad (73)$$

the corresponding substitutions form a group. Its degree is the number of solutions of (73). To count them, first choose α, β not both zero; this gives $p^{2n} - 1$ choices. If a particular solution (γ_0, δ_0) is known, all others have the form

$$\gamma = \gamma_0 + \alpha\xi, \quad \delta = \delta_0 + \beta\xi,$$

with $\xi \in \mathfrak{G}$. Thus there are p^n choices for (γ, δ) , and the degree of the determinant-one group is

$$p^n(p^{2n} - 1). \quad (74)$$

For odd p one can derive a still smaller group. If $p = 2$, a number α is not distinct from $-\alpha$. If p is odd, however, the two matrices

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \quad \begin{pmatrix} -\alpha & -\beta \\ -\gamma & -\delta \end{pmatrix} \quad (75)$$

are distinct, and the whole system splits into such pairs. Multiplication of pairs again gives a pair. Hence these pairs can themselves be regarded as the elements of a new group of degree

$$\frac{1}{2}p^n(p^{2n} - 1). \quad (76)$$

This group, of degree (74) for $p = 2$ and (76) for odd p , will be called the congruence group belonging to $\mathfrak{G}$, and denoted by E . In modern notation it is the projective special linear group over the finite field $\mathfrak{G}$.

For further study it is important to know generators. Write the elements of $\mathfrak{G}$ in the form

$$\xi = x_0 + x_1\varepsilon + \cdots + x_{n-1}\varepsilon^{n-1}, \quad (77)$$

where the x_i are rational residues modulo p . A generating system is

$$A_k = \begin{pmatrix} 1 & \varepsilon^k \\ 0 & 1 \end{pmatrix}, \quad B_k = \begin{pmatrix} 1 & 0 \\ \varepsilon^k & 1 \end{pmatrix}, \quad k = 0, 1, \dots, n-1. \quad (78)$$

Indeed,

$$\begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & \alpha + \beta \\ 0 & 1 \end{pmatrix},$$

and repeated use gives every upper unipotent matrix; similarly every lower unipotent matrix is generated. Combining such elements gives matrices of the forms

$$\begin{pmatrix} \alpha & 1 \\ -1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ \gamma & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and hence, by a further multiplication,

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \quad (79)$$

whenever $\delta \neq 0$. The restriction $\delta \neq 0$ may be removed using the last displayed matrix. Thus the elements (78) generate E .

The group E contains a divisor of index $p^n + 1$, consisting of all substitutions

$$\begin{pmatrix} \alpha & \beta \\ 0 & \delta \end{pmatrix}, \quad \alpha\delta = 1, \quad (80)$$

where β is arbitrary and $\alpha \neq 0$. This divisor yields a permutation representation of E on $p^n + 1$ letters.

The linear expression

$$\eta = \frac{\alpha\xi + \beta}{\gamma\xi + \delta} \quad (81)$$

assigns to every $\xi \in \mathfrak{G}$ a corresponding η , except when $\gamma\xi + \delta = 0$. Conversely, each η determines a ξ , except when $\gamma\eta - \alpha = 0$. To remove the exceptional cases, add to $\mathfrak{G}$ one new element, called infinity and denoted by ∞ . It is governed by the usual rules

$$a + \infty = \infty, \quad a \cdot \infty = \infty \quad (a \neq 0), \quad \frac{a}{0} = \infty, \quad \frac{a}{\infty} = 0.$$

Then every substitution of E acts as a permutation of the $p^n + 1$ symbols of the projective line $\mathfrak{G} \cup \{\infty\}$.

§66. Simplicity of the Group E

The first question in a closer investigation of the group E is whether it has a normal divisor. One can prove that, apart from two quite small exceptional cases, none exists.

Assume that G is a normal divisor of E and that it contains at least one substitution different from the identity,

$$A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}. \quad (82)$$

By normality, for every element $T \in E$ the transform

$$TAT^{-1} \quad (83)$$

also lies in G . We shall show that by such transformations and their products one can derive all elements of E . It is enough to show that the generators A_k, B_k of (78) lie in G .

We proceed step by step.

First, G contains a substitution of the form

$$\begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix}. \quad (84)$$

To see this, form the transform of A by $\begin{pmatrix} 1 & \xi \\ 0 & 1 \end{pmatrix}$. If $\beta \neq 0$, choose ξ so that the lower-left entry vanishes. If $\beta = 0$, then not all of γ and $\delta - \alpha$ vanish, since otherwise A would be the identity; choose ξ so that the transformed matrix has non-zero upper-right entry, and apply the preceding argument once more.

Second, G contains a substitution of the form

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (85)$$

Weber obtains this by solving, for λ, μ, ν, ρ , the system

$$\lambda\rho - \mu\nu = 1, \quad \mu\rho\gamma - \lambda\nu\beta - \lambda\nu\delta = 0, \quad -\mu^2\gamma + \lambda\mu\beta + \lambda\mu\delta = 1. \quad (86)$$

From the last two equations one derives

$$\lambda\beta + \mu\delta = 0, \quad \mu\gamma = \nu, \quad (87)$$

and hence the remaining condition

$$\lambda^2\beta + \lambda\mu\delta - \mu^2\gamma = 1. \quad (88)$$

This equation can always be solved. If β is a square, put $\beta = w^2$ and determine λ . This includes the case $p = 2$. If p is odd and β is a non-square, one reduces the problem to the decomposition of a non-square as a sum of two squares, supplied by §64. Thus the desired transformation exists.

Third, except in the case $n = 1, p = 2$, the group G contains

$$A_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \quad (89)$$

Indeed, from the preceding step the group contains

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and hence also the products

$$\begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & m \end{pmatrix}$$

for every odd integer m . If p is odd, take $m = p$ and obtain A_0 . If $p = 2$, a further calculation gives, for every non-zero $q \in \mathfrak{G}$, a matrix of the form

$$\begin{pmatrix} 0 & q \\ q^{-1} & 0 \end{pmatrix},$$

and products of such matrices yield A_0 provided $n > 1$. The only exception is $n = 1, p = 2$, where one obtains a group of degree six with a normal divisor of degree three.

Fourth, except in the case $n = 1, p = 3$, the group G contains every substitution

$$\begin{pmatrix} 1 & \xi \\ 0 & 1 \end{pmatrix} \quad (90)$$

with arbitrary $\xi \in \mathfrak{G}$. For, using A_0 , one obtains

$$\begin{pmatrix} 1 & (q^2 - 1)\lambda \\ 0 & 1 \end{pmatrix} \quad (91)$$

for every non-zero q . If one can choose q so that $q^2 - 1 \neq 0$, then λ can be adjusted to give any ξ . With a primitive root γ of $\mathfrak{G}$, put $q = \gamma^h$. Such a choice is always possible except in the small cases $n = 1, p = 3$ and $n = 1, p = 5$. The latter is removed by an explicit calculation using the matrix

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -2 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix},$$

which again yields all elements (90).

Thus in all cases except $n = 1, p = 2$ and $n = 1, p = 3$, all generators of E lie in G , hence $G = E$. The group E is therefore simple.

For the exceptional case $n = 1, p = 3$, the group E has degree twelve and contains the normal divisor of degree four

$$G = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} \right\}. \quad (92)$$

The remaining cosets verify directly that this is normal. For $n = 1, p = 2$, the group is isomorphic to the symmetric group on three letters and has its usual normal subgroup of order three.

For the first non-exceptional cases one obtains the degrees

$$n = 1, p = 5 : \quad g = 60,$$

and then the higher degrees supplied by $\frac{1}{2}p^n(p^{2n} - 1)$ for odd p , and by $p^n(p^{2n} - 1)$ for $p = 2$.

§67. Congruence Fields of Degree Two

Every congruence field $\mathfrak{G}_{n,p}$ of degree n and modulus p contains, as a subfield, the field $\mathfrak{G}_{1,p}$ consisting of rational residues modulo p . Consequently the group of linear substitutions $E_{n,p}$ contains as a divisor the group of all substitutions

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad (93)$$

where a, b, c, d are rational residues modulo p and $ad - bc = 1$, again with opposite matrices identified when p is odd.

This group is the group which appears in the theory of elliptic functions as the group of modular equations. It is the real goal of the present investigation. For the study of this group, especially for the search for its divisors, it is most advantageous not to treat it by itself, but to regard it as a divisor of a group $E_{2,p}$.

Let N be a fixed quadratic non-residue modulo p . The polynomial $t^2 - N$ is then irreducible modulo p . We obtain a congruence field of second degree by adjoining a Galois imaginary ε satisfying

$$\varepsilon^2 = N. \quad (94)$$

In what follows, lower-case Latin letters denote rational residues modulo p , which we shall also call real numbers; Greek letters denote elements of the field $\mathfrak{G}_{2,p}$.

The field $\mathfrak{G}_{2,p}$ has the important property that every real number is a square in it. If a is already a quadratic residue modulo p , write $a = x^2$. If b is a non-residue, then $b = aN = (x\varepsilon)^2$ after multiplying by a suitable square a .

A number of the form $a\varepsilon$ will be called purely imaginary. Two numbers $a + b\varepsilon$ and $a - b\varepsilon$ will be called conjugate imaginary. This transfer of terminology from the ordinary complex numbers is justified by the agreement of the calculation rules and cannot cause confusion, since no ordinary complex numbers are involved here.

Since $N^{(p-1)/2} \equiv -1 \pmod{p}$, equation (94) gives

$$\varepsilon^p = -\varepsilon. \quad (95)$$

For $\alpha = a + b\varepsilon$, the binomial theorem gives, modulo p ,

$$\alpha^p = a^p + b^p \varepsilon^p = a - b\varepsilon. \quad (96)$$

Therefore

$$\alpha^{p+1} = a^2 - Nb^2, \quad (97)$$

which is always real.

If γ is a primitive root of $\mathfrak{G}_{2,p}$, then γ^r is real if and only if r is divisible by $p + 1$. In particular γ^{p+1} is a primitive root of the prime field in the ordinary sense.

The necessary and sufficient condition that $\alpha \in \mathfrak{G}_{2,p}$ be real is

$$\alpha^p = \alpha. \quad (98)$$

Moreover every real number is a $(p+1)$ -st power in $\mathfrak{G}_{2,p}$. Conversely, a power γ^r is real only when r is divisible by $p+1$, because every non-zero real number satisfies Fermat's congruence $x^{p-1} = 1$.

The notion of roots of unity also carries over. If m divides $p^2 - 1$ and

$$\rho = \gamma^{(p^2-1)/m},$$

then $\rho^m = 1$, and no lower positive power is one. Thus ρ is a primitive m -th root of unity in $\mathfrak{G}_{2,p}$. All m -th roots of unity are powers ρ^s , and the primitive ones are precisely those with s prime to m . Every non-zero element of $\mathfrak{G}_{2,p}$ is a root of unity of degree $p^2 - 1$.

§68. The Real Linear Congruence Group L_p

Equipped with these tools, we now investigate the real congruence group L_p , consisting of all substitutions

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad ad - bc = 1, \quad (99)$$

where a, b, c, d are real residues modulo p . In the earlier notation this group is $E_{1,p}$. Excluding $p = 2$, its degree is

$$\frac{1}{2}p(p^2 - 1). \quad (100)$$

If A, U are elements of a group, we call

$$U^{-1}AU = A'$$

the element obtained from A by transformation with U . Elements obtained from one another in this way have the same order. If all elements A are taken from a group G , the transformed elements form an isomorphic group $U^{-1}GU$.

Let $A \in L_p$, and let

$$U = \begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix}$$

be an element of $E_{2,p}$. We first try to transform A into the normal form

$$S = \begin{pmatrix} \theta & 0 \\ 0 & \theta^{-1} \end{pmatrix}. \quad (101)$$

From the equation

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix} = \begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix} \begin{pmatrix} \theta & 0 \\ 0 & \theta^{-1} \end{pmatrix}$$

one obtains

$$\begin{aligned} (a - \theta)\lambda + b\nu &= 0, & (a - \theta^{-1})\mu + b\rho &= 0, \\ c\lambda + (d - \theta)\nu &= 0, & c\mu + (d - \theta^{-1})\rho &= 0. \end{aligned} \quad (102)$$

Thus θ and θ^{-1} must be the roots of

$$\theta^2 - \theta(a + d) + 1 = 0. \quad (103)$$

Once θ and θ^{-1} are known, (102) determines the ratios $\nu : \lambda$ and $\rho : \mu$. The scalar factors are still to be chosen so that $\lambda\rho - \mu\nu = 1$. This is possible unless $a + d = \pm 2$, for otherwise the two eigenvectors would be proportional.

The equation (103) is always solvable in $\mathfrak{G}_{2,p}$, since every real number is a square there. Explicitly,

$$\theta = \frac{a+d}{2} \pm \sqrt{\left(\frac{a+d}{2}\right)^2 - 1}. \quad (104)$$

We therefore have:

1. A substitution A is transformable into the normal form S whenever $a + d \neq \pm 2$. The number θ is real or imaginary according as $(a + d)^2 - 4$ is a quadratic residue or a non-residue modulo p .

In the remaining case $a + d = \pm 2$, the normal form S cannot be produced. Instead A can be transformed into

$$T = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad (105)$$

with real t . It is enough to take $a + d = 2$, since the case -2 is obtained by changing all signs. Then

$$a - 1 = -(d - 1), \quad (a - 1)(d - 1) = bc. \quad (106)$$

If $b = 0$ or $c = 0$, the transformation is immediate; if both are non-zero, the displayed identities give an explicit transforming matrix. Hence:

2. A substitution A with $a + d = \pm 2$ can always be transformed into the normal form T .

The orders of elements in L_p are now easily determined. There are three cases.

First kind. If $a + d = \pm 2$, every non-identity substitution is transformed into T . Since

$$T^r = \begin{pmatrix} 1 & rt \\ 0 & 1 \end{pmatrix},$$

such a substitution has order p . Counting these elements gives p^2 substitutions of the first kind, including the identity.

Second kind. If $(a + d)^2 - 4$ is a quadratic residue modulo p , the substitution is transformed into S with real θ . Let γ be a primitive root of $\mathfrak{G}_{2,p}$. Then the real elements may be written as powers $\gamma^{r(p+1)}$, and

$$a + d = \gamma^{r(p+1)} + \gamma^{-r(p+1)}. \quad (107)$$

Distinct values, different from ± 2 , are obtained for

$$r = 1, 2, \dots, \frac{p-3}{2}. \quad (108)$$

The order of S , and hence of the substitution, is a divisor of $(p-1)/2$. It is exactly $(p-1)/2$ when r is prime to $(p-1)/2$. The number of substitutions of the second kind is

$$\frac{1}{2}p(p-3)(p+1). \quad (109)$$

Third kind. If $(a + d)^2 - 4$ is a quadratic non-residue, then θ is not real but is conjugate to its reciprocal. Therefore $\theta^{p+1} = 1$, and the order is a divisor of $(p+1)/2$. Write

$$a + d = \gamma^{r(p-1)} + \gamma^{-r(p-1)}, \quad r = 1, 2, \dots, \frac{p-1}{2}. \quad (110)$$

Here neither $b = 0$ nor $c = 0$ can occur; otherwise a and d would be real eigenvalues. Counting gives

$$\frac{1}{2}p(p-1)(p-1) \quad (111)$$

substitutions of the third kind. Together with the numbers from the first and second kinds, this gives the total degree $\frac{1}{2}p(p^2-1)$, as required.

§69. Imaginary Form of the Group L_p

The substitutions of the first and second kinds in L_p have the property that their corresponding normal forms T or S lie again in L_p , and that they can be transformed into these forms by real substitutions, i.e. by substitutions lying in L_p itself. This does not hold for the substitutions of the third kind.

We shall transform the whole group L_p into another group A_p , which is an isomorphic divisor of $E_{2,p}$, in such a way that the normal forms of the transformations of the third kind are contained in A_p , and each substitution of the third kind in A_p can be transformed into normal form by substitutions of A_p itself.

Consider

$$S = \begin{pmatrix} \theta & 0 \\ 0 & \theta^{-1} \end{pmatrix}, \quad (112)$$

where θ and θ^{-1} are conjugate imaginary. We seek a substitution

$$R = \begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix} \quad (113)$$

so that

$$RSR^{-1} \quad (114)$$

is real. Direct calculation shows that this is achieved when $\lambda\mu$ and $\rho\nu$ are purely imaginary and $\lambda\rho, -\mu\nu$ are conjugate imaginary pairs. A particularly simple choice, with determinant one, is

$$R = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ \varepsilon^{-1} & -\varepsilon^{-1} \end{pmatrix}. \quad (115)$$

If

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is any real substitution in L_p , then, using $\varepsilon^2 = N$, one obtains

$$R^{-1}AR = \begin{pmatrix} \alpha & \beta \\ N\beta' & \alpha' \end{pmatrix}, \quad (116)$$

where α, α' and β, β' are conjugate imaginary pairs; equivalently $\alpha' = \alpha^p, \beta' = \beta^p$. This may also be written in the form

$$\begin{pmatrix} \alpha & \beta \\ N\beta^p & \alpha^p \end{pmatrix}. \quad (117)$$

Conversely, from every matrix of the form (117) one obtains by the inverse transformation a real substitution in L_p . Thus, as A runs through L_p , the transformed matrices form an isomorphic group, which we denote by A_p .

In A_p the normal form S for substitutions of the third kind is present. It remains to check that any substitution A of the third kind in A_p can be brought to this normal form by transformation with a substitution $U \in A_p$. Write

$$U = \begin{pmatrix} \lambda & \mu \\ N\mu' & \lambda' \end{pmatrix}, \quad \lambda\lambda' + N\mu\mu' = 1. \quad (118)$$

The equations corresponding to §68 become

$$(\theta - \alpha)(\theta - \alpha') + N\beta\beta' = 0, \quad (119)$$

and

$$\frac{N\lambda\mu'}{\theta - \alpha} = \frac{\lambda'\mu}{\theta - \alpha'} = \frac{\beta}{\beta'}. \quad (120)$$

The second equation follows from the first by passing to conjugate imaginaries. Hence one can choose

$$N\lambda\mu' = k(\alpha - \theta), \quad N\lambda'\mu = k'(\alpha' - \theta^{-1}),$$

where k, k' are conjugate imaginary, and then determine k, k' so that $\lambda\lambda' + N\mu\mu' = 1$. This proves the claim.

Let γ be a primitive root of $\mathfrak{G}_{2,p}$. For the normal forms of the second and third kinds we may write

$$S_1 = \begin{pmatrix} \gamma^{p+1} & 0 \\ 0 & \gamma^{-(p+1)} \end{pmatrix}, \quad S_2 = \begin{pmatrix} \gamma^{p-1} & 0 \\ 0 & \gamma^{-(p-1)} \end{pmatrix}. \quad (121)$$

If A is a substitution of the second kind in L_p , and $\bar{A}$ a substitution of the third kind in A_p , then there are substitutions $U \in L_p$ and $V \in A_p$ such that

$$A = US_1^r U^{-1}, \quad \bar{A} = VS_2^r V^{-1}.$$

Writing

$$T_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

the same applies to substitutions of the first kind. Hence:

The substitutions of the first, second, and third kinds in L_p or A_p , together with the identity, can be arranged in cycles of respectively

$$p, \quad \frac{p-1}{2}, \quad \frac{p+1}{2}$$

elements.

Two such cycles have no element in common except the identity. For first-kind substitutions this is immediate, since a whole cycle is generated by any non-identity element in it. For the second and third kinds, if two cycles have an element in common, transform one of them so that the common element is in normal form:

$$1, USU^{-1}, US^2U^{-1}, \dots; \quad 1, S, S^2, \dots \quad (122)$$

If for two non-zero exponents

$$US^r U^{-1} = S^s,$$

then calculation forces either U to be diagonal, giving $USU^{-1} = S$, or anti-diagonal, giving $USU^{-1} = S^{-1}$. In both cases the two cycles coincide.

From the enumeration in §68, the numbers of cycles of the first, second, and third kinds are therefore

$$p + 1, \quad \frac{1}{2}p(p + 1), \quad \frac{1}{2}p(p - 1). \quad (123)$$

It is useful, besides L_p and A_p , to consider a third group in $E_{2,p}$, isomorphic to them but not transformable into them within the same field. Since N is real, choose $v \in \mathfrak{G}_{2,p}$ with

$$N = v^{p+1}.$$

To the substitution

$$A = \begin{pmatrix} \alpha & \beta \\ N\beta^p & \alpha^p \end{pmatrix}$$

of A_p let correspond

$$B = \begin{pmatrix} \alpha & v\beta \\ -v^p\beta^p & \alpha^p \end{pmatrix}, \quad \alpha\alpha^p + N\beta\beta^p = 1. \quad (124)$$

The product of two such matrices again has the same form and corresponds to the product of the corresponding A 's. Thus the totality of matrices B forms a group isomorphic to A_p , denoted by F_p .

Replacing $v\beta$ by β gives the simpler form

$$B = \begin{pmatrix} \alpha & \beta \\ -\beta^p & \alpha^p \end{pmatrix}, \quad \alpha\alpha^p + \beta\beta^p = 1. \quad (125)$$

To pass from such a B to the corresponding real substitution A , first form

$$\bar{A} = \begin{pmatrix} \alpha & v^{-1}\beta \\ N(v^{-1}\beta)^p & \alpha^p \end{pmatrix}, \quad (126)$$

and then transform back by

$$A = R\bar{A}R^{-1}. \quad (127)$$

Although F_p is isomorphic to L_p and A_p , it cannot be transformed into either by substitutions whose entries belong to $\mathfrak{G}_{2,p}$. To perform such a transformation one would have to pass to a higher field in which all elements of $\mathfrak{G}_{2,p}$ become squares.